

SUMERIX INTERNSHIP PROJECT DOCUMENTATION

Responsive Personal Portfolio & Business Landing Website

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DOMAIN: FULL STACK WEB DEVELOPMENT

TASK: 1 (FINAL PRODUCTION RELEASE)



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SUMERIX GROBAL

1. PROJECT INTRODUCTION DETAILS

Context:

In the discipline of professional Full stack Web Development, rigorous project documentation is the definitive bridge between raw code and a maintainable product. High-quality documentation is a strategic asset that establishes technical accountability and ensures that architectural decisions remain transparent over time. By articulating the design patterns, engineering logic, and compliance standards utilized, provides the necessary roadmap for future scalability, distinguishing an enterprise-grade application from a transient prototype.

Title Page Details:

Attribute	Specification Details
Project Title	Responsive Personal Portfolio & Business Landing Web Application
Author	Anjan Basak
Internship Domain	Full Stack Web Development
Task Module	Task 1 of 6 (Web Development Fundamentals)
Document Version	1.0, September 2026
Status	Final Production Release

Certificate of Authenticity

OFFICIAL CERTIFICATION

This is to certify that the project report entitled "Responsive Personal Portfolio & Business Landing Web Application" represents an authentic record of frontend development, semantic markup, responsive UI design, client-side scripting, and accessibility implementation. This work, completed by Anjan Basak, has been verified for cross-browser compliance (Chrome v90+, Firefox v88+), mobile-first responsiveness, and strict adherence to WCAG/ARIA accessibility standards.

Anjan Basak

Frontend Developer & Technical Lead — September 2026

Declaration

I hereby declare that this project report entitled "Responsive Personal Portfolio & Business Landing Web Application" is an authentic record of my own web engineering work. All source code, semantic HTML5 structure, HSL-based CSS3 layout architecture, client-side JavaScript execution, and version control histories presented herein were authored and configured directly by me in accordance with Task-1 industry guidelines and Sumerix Global standards.

Anjan Basak

Kolkata, India — September 2026

Acknowledgements

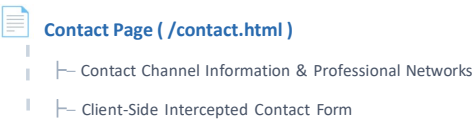
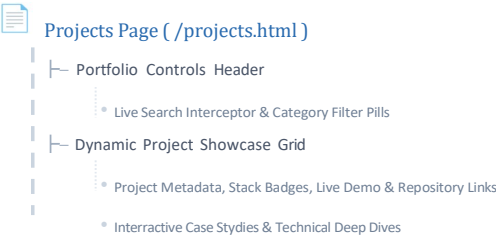
Professional gratitude is extended to the W3C for standard specifications and the WCAG/ARIA documentation. Technical appreciation is directed toward the maintainers of MDN Web Docs and freeCodeCamp for providing the reliable educational resources that informed this architecture. Furthermore, I acknowledge the open-source community, specifically the Git and GitHub Pages infrastructure teams, for providing the deployment pipelines essential for public project delivery.

Abstract

The Responsive Personal Portfolio & Business Landing Web Application serves as a foundational frontend engineering platform, showcasing a transition from core markup to a sophisticated multi-page system. Built on a "Zero-Dependency" architecture, the project utilizes semantic HTML5, modular CSS3, and ES6+ JavaScript. The platform facilitates a professional showcase featuring live metrics, testimonials, and high-fidelity project cards—including clones of Amazon, Spotify, and Zomato. Key technical achievements include a persistent theme engine using the Web Storage API, asynchronous form validation, and WCAG-compliant navigation systems, all optimized for the Critical Rendering Path.

1. WEBSITE SITEMAP & ROUTE STRUCTURE

The following hierarchical breakdown outlines the complete routing, page modules, and functional components engineered across the application platform.



2. SOFTWARE DEVELOPMENT ROADMAP

Planned engineering milestones outlining completed implementation phases and upcoming technical releases.

Phase 1: Core Foundation & UI Framework

✓ COMPLETED

- Engineered responsive mobile navigation drawer and dark mode state engine with LocalStorage persistence.
- Configured global CSS design tokens, custom theme variables, and modular component stylesheets.

Phase 2: Interactive Modules & Form Validation

✓ COMPLETED

- Implemented Regex-based client-side form interception with real-time field error handling.
- Integrated dynamic project showcase grid with interactive metadata filtering and modal views.

Phase 3: QA Testing, Automation & Performance

🕒 IN PROGRESS

- Execute automated cross-browser unit tests and optimize assets to achieve 95+ Lighthouse performance scores.
- Refactor database queries and API routing for low-latency client requests.

Phase 4: Internationalization (i18n) & PWA Capabilities

📅 17 UPCOMING

- Integrate multi-language locale switcher for global accessibility.
- Register custom Service Worker for offline asset caching and progressive web app (PWA) installation.

Implementation Note: Both the sitemap hierarchy and software roadmap are maintained synchronously with the main repository documentation.

3.PROJECT INTRODUCTION & CORE MOTIVATIONS

Context & Strategic Importance

The modern web ecosystem is plagued by "framework bloat," where heavy Node.js dependency chains and complex bundlers create significant maintenance overhead and security risks. Adopting a "framework-free" paradigm is a strategic engineering choice aimed at maximizing performance. By executing code directly in the browser's native engine, we eliminate the instability of the modern build-chain and ensure the application remains functional regardless of the shifting library landscape.

Core Engineering Principles

- **Zero-Framework Architecture:** Eliminating build-tool fatigue and transpilation steps (Babel, Webpack) to ensure zero-vulnerability audits and instant time-to-market.
- **Maintainable Design Systems:** Leveraging HSL-based CSS Custom Properties for programmatic control over typography, spacing, and theming.
- **Accessible UI Componentry:** Utilizing ARIA landmark roles and dynamic states (aria-expanded) to ensure full screen-reader and keyboard compatibility.
- **Client-Side State Persistence:** Using the browser's Storage API to retain user preferences without the latency of a backend round-trip.

Mission Analysis

The "Zero-Dependency" approach effectively mitigates package deprecation risks and security vulnerabilities inherent in third-party npm modules. By bypassing the transpilation layer, we reduce the complexity of the deployment pipeline and ensure that the source code remains an accurate representation of the runtime execution, facilitating easier debugging and auditability.

4.COMPARATIVE SYSTEM ANALYSIS: TRADITIONAL VS. ROPOSED

Context & Strategic Importance

The evolution of professional identity has shifted from static documents to dynamic, self-hosted web presences. For a systems architect, a custom-coded solution is superior to proprietary abstractions, as it demonstrates total ownership of the stack and a deep understanding of the underlying browser APIs.

Gap Analysis

Feature	Traditional (PDF/Wix)	Systems Proposed Core Web Standard System
Interactivity	Static or limited by template.	Dynamic DOM manipulation (Drawers, Themes).
Code Quality	Unverifiable / Bloated.	Hand-crafted, semantic, and high-performance.
Performance	High script overhead.	Zero-dependency, optimized Critical Path.
Accessibility	Frequently omits ARIA.	Native WCAG/ARIA compliance.
User Experience	Non-responsive PDF zoom.	Mobile-first, fluid responsive grids.

Evaluating Impacts

Traditional systems suffer from Bloated Asset Overhead, where automated site builders inject unnecessary libraries that degrade mobile Core Web Vitals. Furthermore, they result in Unverifiable Source Code Quality, preventing recruiters from assessing a developer's ability to implement clean hierarchy. The proposed system provides Verifiable Proof of Competence by exposing a clean, professional codebase.

System Capabilities

- Multi-Page Semantic Structure: Organized modules (about, services, projects, contact) with consistent global landmarks.
- Dynamic Interactive Showcase: Integration of complex clones (Spotify, Amazon) and browser games within a unified UI.

5. LOCAL DEVELOPMENT ENVIRONMENT SETUP

Context & Strategic Importance

A standardized environment ensures that the development experience mirrors production as closely as possible. Rigorous documentation of the toolchain is essential for maintaining cross-browser reliability and consistent code formatting across the lifecycle.

Toolchain Specifications

- Modern Browser Engines: Google Chrome (v90+) for performance auditing; Mozilla Firefox (v88+) for advanced CSS Grid debugging.
- IDE: Visual Studio Code (VS Code) with 2-space indentation enforcement.
- Quality Tools: Prettier for automated style normalization and Chrome DevTools for accessibility auditing.

Workflow Optimization

The environment leverages the VS Code Live Server for instant hot-reloading. Version control is managed strictly via Git CLI, utilizing feature branches to ensure the integrity of the main production branch.

6.TECHNICAL STACK SPECIFICATIONS

Context & Strategic Importance

The "Markup, Style, and Script" triad forms the foundation of modern high-performance web applications. This stack selection prioritizes native browser APIs to achieve the lowest possible "Time to Interactive" (TTI).

Layered Breakdown

Layer	Technology	Strategic Implementation
Markup	HTML5	Semantic landmarks, ARIA roles, native form controls.
Styling	CSS3	Flexbox, CSS Grid, HSL Token Architecture.
Scripting	ES6+ JS	DOM Manipulation, Web Storage, matchMedia API.
Deployment	Static Hosting	GitHub Pages (CDN-backed static engine).

Performance Transformation

By choosing a dependency-free stack, the application achieves near-perfect Google Lighthouse scores. The use of HSL tokens rather than Hex codes allows for the programmatic generation of tints and shades by merely adjusting the lightness value, providing a more robust design system for future theme extensions.

7.SYSTEM ARCHITECTURE & FILE HIERARCHY

Context & Strategic Importance

A modular directory structure is a prerequisite for scalability. By isolating concerns into dedicated folders, we ensure that the codebase is navigable and that asset dependencies are explicitly clear, facilitating rapid onboarding and maintenance.

Modular Tree Representation

```
portfolio-website/
├── assets/
│   ├── css/
│   │   ├── style.css      # Design tokens, variables & base resets
│   │   └── responsive.css # Media queries & fluid
│   ├── js/
│   │   ├── script.js      # Theme engine & persistence
│   │   ├── navigation.js  # Accessible drawer logic #
│   │   └── validation.js  # Form validation engine
│   ├── images/            # Optimized assets (Amazon/Spotify previews) #
│   └── icons/             # Vector SVG UI iconography
├── pages/
│   ├── about.html         # Bio, skills matrix & metrics
│   ├── services.html      # Service tiers & architecture consulting
│   ├── projects.html      # Portfolio grid (Clones & Games)
│   ├── contact.html       # Validation-enabled communication portal
│   └── index.html         # Entry point (Hero, Metrics, Testimonials)
```

Hub-and-Spoke Navigation Map



Relative Pathing Logic : A critical engineering requirement is the use of relative stepping logic. Files in the /pages/ directory must utilize the ../assets/ prefix to resolve resources correctly across different directory depths, preventing broken links in production subdirectories.

8. CLIENT-SIDE SCRIPTING: ACCESSIBLE NAVIGATION SYSTEM

Context & Strategic Importance

Navigation is the primary point of failure for mobile accessibility. This module implements a robust "Dynamic Injection" pattern, ensuring the hamburger menu is accessible to screen readers while managing viewport scroll-locking to maintain UX integrity.

Full Module Implementation (navigation.js)

```
/* --- Accessible Hamburger Mobile Navigation System --- */ document.addEventListener("DOMContentLoaded", () => {
  const navbar = document.querySelector(".navbar");
  const navLinks = document.getElementById("navLinks") || document.querySelector(".nav-links");
  // Dynamic Button Injection
  if (navbar && !document.getElementById("hamburger")) { navbar.insertAdjacentHTML(
    "beforeend",
    `<button class="hamburger" id="hamburger" aria-label="Toggle Menu" aria-
    expanded="false" aria-controls="navLinks">
      
      `
  );
  }
  const hamburger = document.getElementById("hamburger");
  const navItems = navLinks ? navLinks.querySelectorAll("a") : [];

  if (hamburger && navLinks) {
    const toggleMenu = (isOpen) => {
      const activeState = typeof isOpen === "boolean" ? isOpen :
!hamburger.classList.contains("active"); hamburger.classList.toggle("active",
      activeState); navLinks.classList.toggle("active", activeState);
      hamburger.setAttribute("aria-expanded", activeState ? "true" : "false");
      document.body.classList.toggle("no-scroll", activeState);
    };

    hamburger.addEventListener("click", toggleMenu); navItems.forEach((link) =>
    link.addEventListener("click", () =>
toggleMenu(false)));

    document.addEventListener("keydown", (e) => {
      if (e.key === "Escape" && hamburger.classList.contains("active")) { toggleMenu(false);
        hamburger.focus();
      }
    });
  }
}
```

Analytical Deep Dive : The script uses dynamic icon toggling and state-driven ARIA updates. The use of `document.body.classList.toggle("no-scroll")` is critical for preventing "background scroll leak," a common UX failure in mobile navigation drawers.

9.CLIENT-SIDE SCRIPTING: PERSISTENT THEME ENGINE

Context & Strategic Importance

Dark mode implementation requires careful management of the "theme-flicker." By executing the script as a blocking resource prior to the First Contentful Paint (FCP), we ensure that the document root is styled before the user sees the content.

Full Module Implementation (script.js)

```
/* --- Theme Manager & Preference Listener --- */
const SELECTORS = { themeToggleSwitch: "#themeToggle" };

function initThemeManager() {
  const toggleSwitch = document.querySelector(SELECTORS.themeToggleSwitch); const currentTheme =
  localStorage.getItem("theme");
  const systemPrefersDark = window.matchMedia("(prefers-color-scheme: dark)").matches;

  // Initial State Check if
  (currentTheme) {
    document.documentElement.setAttribute("data-theme", currentTheme); if (toggleSwitch)
    toggleSwitch.checked = (currentTheme === "dark");
  } else if (systemPrefersDark) { document.documentElement.setAttribute("data-
  theme", "dark"); if (toggleSwitch) toggleSwitch.checked = true;
  }

  // Manual Toggle Listener if
  (toggleSwitch) {
    toggleSwitch.addEventListener("change", (e) => {
      const newTheme = e.target.checked ? "dark" : "light";
      document.documentElement.setAttribute("data-theme", newTheme);
      localStorage.setItem("theme", newTheme);
    });
  }

  // System Preference Listener
  window.matchMedia("(prefers-color-scheme: dark)").addEventListener("change", (e) => { if
  (!localStorage.getItem("theme")) {
    const newTheme = e.matches ? "dark" : "light";
    document.documentElement.setAttribute("data-theme", newTheme);
  }
  });
}
document.addEventListener("DOMContentLoaded", initThemeManager);
```

Logic Analysis

The inclusion of the `window.matchMedia().addEventListener` ensures that if a user changes their OS theme while the application is open, the UI updates dynamically without requiring a manual reload or a `localStorage` override.

10. CLIENT-SIDE SCRIPTING: CLIENT-SIDE FORM VALIDATION

Context & Strategic Importance

Client-side validation is a UX optimization that prevents the latency of a server round-trip for malformed data. This module uses Regular Expressions and character length thresholds to provide instantaneous feedback.

Full Module Implementation (validation.js)

```
/* --- Form Interceptor & Regex Validation --- */
document.addEventListener("DOMContentLoaded", () => {
  const contactForm = document.getElementById("contactForm");
  const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

  if (contactForm) {
    contactForm.addEventListener("submit", (e) => {
      e.preventDefault();
      let isValid = true;

      const name = document.getElementById("nameInput");
      const email = document.getElementById("emailInput");
      const phone = document.getElementById("phoneInput");
      const msg = document.getElementById("messageInput");

      if (name.value.trim().length < 2) isValid = false;
      if (!emailRegex.test(email.value.trim())) isValid = false;
      if (phone.value.trim().replace(/\D/g, "").length !== 10) isValid = false;
      if (msg.value.trim().length < 10) isValid = false;

      if (isValid) {
        document.getElementById("successMessage").style.display = "block";
        contactForm.reset();
      } else {
        alert("Please ensure all fields are filled correctly.");
      }
    });
  }
});
```

11. GLOBAL DESIGN SYSTEM & HSL TOKEN ARCHITECTURE

Context & Strategic Importance

CSS Custom Properties (tokens) are used to centralize branding. The choice of HSL (Hue, Saturation, Lightness) over Hex or RGB is strategic: it allows the architect to create an entire color system by simply modifying the `--primary-hue` variable.

Token Architecture

```
:root {  
  /* Brand Palette - Programmable HSL */  
  --primary-hue: 262;  
  --accent-color: hsl(var(--primary-hue), 83%, 58%);  
  --accent-hover: hsl(var(--primary-hue), 83%, 48%);  
  
  /* Light Theme Tokens */  
  --bg-color: #f8fafc;  
  --card-bg: #ffffff;  
  --text-main: #0f172a;  
  --border-color: #e2e8f0;  
}  
  
[data-theme="dark"] {  
  /* Dark Theme Tokens */  
  --bg-color: #0f172a;  
  --card-bg: #1e293b;  
  --text-main: #f8fafc;  
  --border-color: #334155;  
}
```

Dark-Theme Overrides

In dark mode, image assets (such as navigation icons) utilize `filter: brightness(0) invert(1)` to ensure high contrast without requiring multiple asset downloads, further optimizing the network payload.

12. RESPONSIVE VIEWPORT BREAKPOINT STRATEGIES

Context & Strategic Importance

A mobile-first philosophy ensures the smallest devices receive the most optimized experience. The architecture uses fluid grids to scale gracefully as screen real estate increases.

Breakpoint Categorization

- Mobile (< 640px): Single-column stacks; full-width buttons; off-canvas navigation drawer.
- Tablet (640px–1024px): 2-column project grids; horizontal pill navigation; increased padding.
- Desktop (> 1024px): 3-column project grids; expanded metadata; persistent sidebar elements.

13. STATIC PRODUCTION DEPLOYMENT

Context & Strategic Importance

Deployment is the final phase of the engineering lifecycle. Using GitHub Pages provides a secure, CDN-backed environment that ensures low latency for global visitors.

Deployment Pipelines

The project is hosted via GitHub Pages, with manual deployment via the main branch. Automation can be extended via GitHub Actions to validate markup and CSS before the site is updated.

Quality Assurance & Semantic Git

Commits are managed using "Conventional Commits":

- feat: Adding a new clone (e.g., Zomato Clone).
- fix: Resolving an ARIA state error.
- style: Updating the HSL primary hue.

Interactive Developer FAQ

Question	Technical Solution / Architecture Response
How do I update the primary brand color?	Change the --primary-hue variable in style.css.
Why use relative paths?	Absolute paths break when hosted in subfolders. See Section 6 for stepping logic.
Is Node.js required?	No, the architecture is entirely zero-dependency and framework-free.
How do I fix broken images in subpages?	Ensure you use ../assets/ to step out of the /pages/ folder.
How do I add a new project?	Duplicate a project card in projects.html and update the metadata.

14. CONCLUSION

The zero-dependency architecture implemented for Task-1 proves that professional, high-performance web platforms can be delivered through the mastery of native browser APIs. By rejecting framework overhead, the application achieves superior Core Web Vitals, seamless accessibility, and persistent state management. This project serves as a robust foundation for a scalable, standards-compliant portfolio system.